Minibatch Heavy-Ball SGD with Mode-Dependent Kurtosis: Threshold, Exact Rate, Prefactor and Critical Slowing Down

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Abstract

For heavy-ball SGD on a quadratic in $\mathbb{R}^P$ with minibatch Hessians H_{B_t} of mean $H \succ 0$, we study the expected loss at finite dimension P and finite batch B when the features have independent coordinates in the eigenbasis of H, with a fourth moment that may differ from mode to mode, possibly multiplied by a common random scale (per-coordinate Bernoulli gates, a per-sample gate on an arbitrary H, Student-t features). When the fourth moments differ across modes the curvature noise is not of the spectrally expressible form of earlier generating-function analyses. We show that the second-moment operator $\mathcal{T}$ has the spectral radius of its restriction to diagonal blocks at every step size, that the mean-square threshold is the root of a rank-one secular equation, that twice the expected loss obeys an exact discrete renewal equation, and that $\rho(\mathcal{T}) = 1/z_\star$, where $z_\star$ is the smallest positive root of $z\widehat{K}(z) = 1$ for the generating function $\widehat{K}$ of the renewal kernel. When $\gamma(2+\kappa_k) \geq 1$ for every mode k (for instance $\kappa_k \geq -1$), where γ is the fourth moment of the random scale and $3+\kappa_k$ that of mode k, the loss approaches its noise floor as a single exponential $C\rho(\mathcal{T})^t$ with an explicit $C > 0$, and $1-\rho(\mathcal{T})$ vanishes linearly at the threshold with an explicit slope. For Gaussian features the secular equation [Ziyin et al., 2022, Wang et al., 2026], the renewal equation [Wang et al., 2026] and the rate as a root of the same equation, with a prefactor in the divergent regime [Velikanov et al., 2023], are known; for isotropic gated features the threshold and the renewal equation are in Everett and Paquette [2026]; the general threshold is in Cai et al. [2026]. Consequences: a one-direction stability test can overestimate the stable step by two orders of magnitude while the true threshold changes by about 1% between Gaussian and Rademacher features. In the EMA form of momentum the secular sum depends on β only through $(1-\beta)/(1+\beta)$, so the threshold and every level set of S increase with β, by an amount between $\frac{\beta}{1+\beta}\lambda_{\min}$ and $\frac{\beta}{1+\beta}\lambda_{\max}$ in $1/\eta_{\mathrm{EMA}}$, with a finite limit as $\beta \to 1$ for $P \geq 2$, and the ratio of the threshold to the deterministic one depends on (β, B) only through $B(1-\beta)/(1+\beta)$. With label noise a fixed loss floor is a level set of S, the rate satisfies $1-\rho(\mathcal{T}) \leq 1-S$, and the speedup of momentum at a fixed floor ranges from none (one direction, high floor) to above the factor 2 found for a single isotropic feature (anisotropic H, moderate floor). In the general setting, at fixed effective step $\eta/(1-\beta)$ momentum strictly raises the threshold and lowers $\mathrm{tr}(\Sigma_\xi \Sigma_0)$, but can raise the loss.

1 Introduction

Consider heavy-ball SGD on a quadratic loss in $\mathbb{R}^P$ with mean Hessian $H = \sum_{k=1}^P \lambda_k u_k u_k^\top \succ 0$, $\lambda_1 \geq \cdots \geq \lambda_P > 0$:

$$w_{t+1} = \big((1+\beta)I - \eta H_{B_t}\big)w_t - \beta w_{t-1} - \eta\xi_t, \qquad \beta \in [0,1),\ \eta > 0, \tag{1}$$

where (H_{B_t}, ξ_t) are i.i.d. across steps and independent of (w_0, w_{-1}), with $\mathbb{E}H_{B_t} = H$, $\mathbb{E}\|H_{B_t}\|_F^2 < \infty$, $\mathbb{E}\xi_t = 0$ and $\Sigma_\xi = \mathbb{E}\xi_t\xi_t^\top \succ 0$. Nothing is assumed on the joint law of H_{B_t} and ξ_t within a step. In

the EMA form of momentum, $m_{t+1} = \beta m_t + (1-\beta) g_t$, $w_{t+1} = w_t - \eta_{\mathrm{EMA}} m_{t+1}$, the iterates are those of (1) with $\eta = (1-\beta)\eta_{\mathrm{EMA}}$. Write $s_t = (w_t, w_{t-1})$, $Q_t = \mathbb{E} s_t s_t^\top$, $F_t = \begin{pmatrix} (1+\beta)I - \eta H_{B_t} & -\beta I \\ I & 0 \end{pmatrix}$, $\bar{F} = \mathbb{E} F_t$, $\mathcal{V}(X) = \mathbb{E}[(H_{B_t} - H) X (H_{B_t} - H)]$ and $\mathcal{T}(Q) = \mathbb{E}[F_t Q F_t^\top]$.

What is known. *Stability.* With non-degenerate additive noise, bounded second moments are equivalent to $\rho(\mathcal{T}) < 1$ [Costa et al., 2005, Ch. 3]. For LMS with Gaussian inputs ($B = 1$, $\beta = 0$) the recursion of the diagonal second moments is a diagonal plus rank-one matrix, and its eigenvalues are the roots of the corresponding secular equation [Feuer and Weinstein, 1985, eqs. (20) and (23)]; for momentum LMS the characteristic polynomial has a factor $Q(z)$ of the form $1 - z\widehat{K}(z)$ below [Roy and Shynk, 1990, Section IV]. For SGD at interpolating minima the exact threshold is Mulayoff and Michaeli [2024, Theorem 5]; for heavy ball and arbitrary non-commuting minibatch Hessians it is Cai et al. [2026, Theorem F.1] (Proposition 3 below), whose Lemma F.13 contains the operator of Theorem 2(2). The stationary covariance equation with momentum, under ergodicity, is due to Liu et al. [2021] (Theorem 1 of Ziyin et al. [2022]); for Gaussian features and label noise Ziyin et al. [2022, Theorem 2] solve it in closed form, and its denominator is the secular equation of Theorem 5 at $\kappa_k \equiv 0$.

Loss dynamics. Without momentum, Bordelon and Pehlevan [2022, Theorem 1] give the exact loss for Gaussian features through the matrix $\mathcal{T}_{\mathrm{diag}}$ below at $\beta = 0$, and their Theorem 2 gives it for arbitrary features as the iterate of the P^2-dimensional second-moment recursion, whose coefficients are the fourth-moment tensor (their eq. (48)), without additive noise and without a threshold or rate formula. For Gaussian features with momentum and batch, Velikanov et al. [2023] write the loss generating function in closed form (their spectrally expressible approximation is exact there), state stability as $\widetilde{U}(1) < 1$, and in the divergent regime obtain the rate from $r\widetilde{U}(r) = 1$ and a prefactor under an exponential ansatz (their eq. (24)); Wang et al. [2026, Appendix B] write the loss as a renewal equation, obtain the threshold from its kernel mass (their eq. (56)) and bound it (their eq. (57)). For isotropic least squares with Bernoulli-gated Gaussian inputs, Everett and Paquette [2026] give an exact discrete renewal equation at every d and B (their Lemma C.1), the threshold (their eq. (6)), the floor and geometric convergence (their Corollary C.2) and the per-step spectrum through a cubic (their Lemma H.4); their vocabulary model has rows that decouple (their Lemma H.12), they note that an anisotropic H gives one coupled block per eigenvalue (their Remark B.2), and they propose anisotropic features as future work. In high-dimensional limits the loss of SGD obeys a Volterra equation whose Malthusian root gives the rate, with an explicit prefactor [Paquette et al., 2021, Theorem E.2]; with momentum, Paquette and Paquette [2021] find that small-batch heavy ball gives no improvement over SGD once the step is adjusted, and Lee et al. [2022] study the interpolating case.

One-direction tests. The curvature variance along the top eigenvector gives a necessary condition for stability [Mulayoff and Michaeli, 2024, Proposition 9]; batch sharpness above $2/\eta$ certifies instability, and the criterion is one-sided by design [Andreyev and Beneventano, 2024, Theorem 1]; Andreyev et al. [2026, eq. (10)] give the one-dimensional threshold with momentum.

What this note adds.

1. A model beyond the spectrally expressible class: independent coordinates with mode-dependent fourth moments and a common random scale. For it, $\rho(\mathcal{T}) = \rho(\mathcal{T}_{\mathrm{diag}})$ at every η, the secular threshold and the exact renewal equation (Lemma 4, Theorem 5(a),(b)). It contains the Gaussian case and the isotropic gated case of Everett and Paquette [2026], and extends the latter to an arbitrary H (Corollary 6).

2. A proof that $\rho(\mathcal{T}) = 1/z_\star$ at finite P for every η (Theorem 5(c)); on the stable side with additive noise, the single-exponential approach to the floor with an explicit $C > 0$, and the slope of

critical slowing down at the threshold (Theorem 5(d),(e)). We did not find the last two in prior work, also for Gaussian features.

3. The divergence half of the stability criterion, with linear growth from every initialization and without any assumption on the joint law of (H_{B_t}, ξ_t) (Lemma 1; the boundedness half with correlated forcing is Cai et al. [2026, Proposition C.3]), and the step-covariance identity $\mathbb{E}[\Delta w \Delta w^\top] = \frac{\eta}{1+\beta}(H\Sigma_0 + \Sigma_0 H)$ (Theorem 2(1)).

4. Momentum and batch in EMA form (Corollary 6(iv)): monotonicity of every level set of S in β with two-sided bounds, the limit $\beta \to 1$, and the dependence of $\eta^\star/\eta_{\mathrm{det}}$ on $B(1-\beta)/(1+\beta)$ alone; with label noise, the rate at a fixed loss floor (Proposition 9), extending the isotropic comparisons of Everett and Paquette [2026, Proposition H.10, Lemma C.5].

5. The size of the gap between the one-direction threshold and the true threshold for non-Gaussian features, against a near-universality of the true threshold in the kurtosis (Proposition 7; for Gaussian features Wang et al. [2026, eq. (57)] give the same upper bound and the lower bounds $\alpha\lambda_1$ and $p\kappa_\beta \operatorname{tr} H$), and, in the general setting, the strict increase of $\eta^\star/(1-\beta)$ in β with an explicit margin and a signed finite-step comparison at fixed effective step (Proposition 8; the monotonicity is eq. (67) of Cai et al. [2026], written there as a non-strict inequality, and the margin is new).

Proofs are in Appendix A; every number is printed by `paper/checks/note-v2-checks.py`.

2 General facts

Lemma 1 (Stability without assumptions on the joint law)**.** *Assume the setting of (1) and that (w_0, w_{-1}) has finite second moments. Let $G = \mathrm{blkdiag}(\eta^2\Sigma_\xi, 0)$.*

(i) $\mathcal{T}$ preserves the symmetric and the positive semidefinite matrices, $\|\mathcal{T}^k\| = \|\mathcal{T}^k(I)\|$ in the operator norm induced by the spectral norm, and $\rho(\mathcal{T}) = \rho(\mathcal{T}|_{\mathbb{S}^{2P}})$.

(ii) If $\rho(\mathcal{T}) < 1$, then $Q_t \to Q_\infty = \sum_{k\geq 0} \mathcal{T}^k(G)$ and $\mathbb{E}w_t \to 0$, whatever the joint law of (H_{B_t}, ξ_t).

(iii) If $\rho(\mathcal{T}) \geq 1$, there is $c > 0$ with $\mathbb{E}\|w_t\|^2 + \mathbb{E}\|w_{t-1}\|^2 \geq ct$ for every $t \geq 1$.

The assumption $H \succ 0$ cannot be dropped in (iii): for $P = 1$, $H = 0$, $H_{B_t} = \epsilon_t \in \{\pm 1\}$ equiprobable, $\xi_t = -\epsilon_t$ and $w_0 = w_{-1} = 1$, one has $w_t \equiv 1$ while $\rho(\mathcal{T}) \geq \rho(\bar{F})^2 = 1$.

Part (ii), with a forcing correlated with the transition, is Cai et al. [2026, Proposition C.3], and their Proposition A.3 gives, above their threshold, an unbounded second moment from some deterministic initialization; (iii) gives linear growth from every initialization.

Theorem 2 (Stationary covariance equation)**.** *Let $\rho(\mathcal{T}) < 1$ and let $\Sigma_0 = \mathbb{E}w_t w_t^\top$, $D = \mathbb{E}[(w_{t+1} - w_t)(w_{t+1} - w_t)^\top]$ in the stationary regime of Lemma 1(ii). Then*

(1) $D = \frac{\eta}{1+\beta}(H\Sigma_0 + \Sigma_0 H)$*;*

(2) $$(H\Sigma_0 + \Sigma_0 H) - \frac{\eta}{1+\beta} H\Sigma_0 H + \frac{\eta\beta}{(1-\beta)(1-\beta^2)}[H,[H,\Sigma_0]] - \frac{\eta}{1-\beta}\mathcal{V}(\Sigma_0) = \frac{\eta}{1-\beta}\Sigma_\xi.$$

With $\mathcal{V}$ replaced by a given noise covariance, (2) is the equation of Liu et al. [2021] (Theorem 1 of Ziyin et al. [2022]), written with $H^2\Sigma_0 + \Sigma_0 H^2 = [H,[H,\Sigma_0]] + 2H\Sigma_0 H$. Write (2) as $(\mathcal{L}_0 - \eta\mathcal{B}_\beta)\Sigma_0 = \frac{\eta}{1-\beta}\Sigma_\xi$ with $\mathcal{L}_0 X = HX + XH$ and $\mathcal{B}_\beta X = \frac{1}{1+\beta} HXH - \frac{\beta}{(1-\beta)(1-\beta^2)}[H,[H,X]] + \frac{1}{1-\beta}\mathcal{V}(X)$, both self-adjoint on $(\mathbb{S}^P, \langle X, Y\rangle = \operatorname{tr} XY)$.

Proposition 3 (Cai et al., 2026, Theorem F.1)**.** *$\rho(\mathcal{T}) < 1$ if and only if $\mathcal{L}_0 - \eta\mathcal{B}_\beta \succ 0$ on $\mathbb{S}^P$, i.e. $\eta < \eta^\star \triangleq 1/\lambda_{\max}(\mathcal{L}_0^{-1/2}\mathcal{B}_\beta\mathcal{L}_0^{-1/2})$.*

Cai et al. [2026] state it as $(1-b)/\alpha > r_C(b)$ for a quotient $q_{C,b}$ maximized over positive semidefinite X. The maximum over all of $\mathbb{S}^P$ is the same: in their notation $D^{-1}\mathcal{V}$ preserves the positive semidefinite cone [Cai et al., 2026, Lemma F.13], so by the Krein–Rutman theorem its spectral radius is an eigenvalue with a positive semidefinite eigenmatrix, and $D^{-1}\mathcal{V}$ is similar to the self-adjoint $D^{-1/2}\mathcal{V}D^{-1/2}$, whose top eigenvalue is therefore that spectral radius. At $\beta = 0$, under the sampling and positive-semidefinite sample-Hessian assumptions of Mulayoff and Michaeli [2024, Theorem 5], it recovers their threshold.

3 Independent coordinates with mode-dependent fourth moments

Let $H_{B_t} = \frac{1}{B}\sum_{b=1}^{B} z_{t,b}z_{t,b}^\top$ with i.i.d. $z_{t,b} = \zeta_{t,b}H^{1/2}\omega_{t,b}$, where the scalar ζ is independent of ω, $\mathbb{E}\zeta^2 = 1$, $\gamma \triangleq \mathbb{E}\zeta^4$ (≥ 1), and the coordinates $\omega_k = u_k^\top\omega$ are independent with $\mathbb{E}\omega_k = 0$, $\mathbb{E}\omega_k^2 = 1$, $\mathbb{E}\omega_k^4 = 3 + \kappa_k$. Put $p = 1/B$, $\kappa_\beta = \frac{1+\beta}{1-\beta}$, and for each mode

$$c_k \triangleq p\big(\gamma(2+\kappa_k) - 1\big)\lambda_k^2, \qquad \alpha_k \triangleq 1 + \kappa_\beta\frac{c_k}{\lambda_k^2}, \qquad x_k \triangleq \frac{\eta\lambda_k}{2(1+\beta)}, \qquad a_k \triangleq 1 + \beta - \eta\lambda_k.$$

Examples: (a) $\zeta \equiv 1$, $\kappa_k \equiv \kappa$ (Gaussian $\kappa = 0$, Rademacher $\kappa = -2$); (b) per-coordinate gates $\omega_k = (s_k/\sqrt{q_k})\tilde{\omega}_k$, $s_k \sim$ Bernoulli(q_k) independent, so $\kappa_k = (3 + \tilde{\kappa}_k)/q_k - 3$ and λ_k carries the frequency q_k; (c) the per-sample gate $\zeta = s/\sqrt{q}$, $s \sim$ Bernoulli(q), of Everett and Paquette [2026], now on an arbitrary H: $\gamma = 1/q$; (d) Student-t_ν features, $\nu > 4$: $\gamma = \frac{\nu-2}{\nu-4}$, $\kappa_k = 0$.

Lemma 4 (Curvature noise and diagonal reduction)**.** *In the eigenbasis of H, for $X \in \mathbb{S}^P$,*

$$\mathcal{V}(X)_{kk} = p\gamma\lambda_k\,\mathrm{tr}(HX) + c_kX_{kk}, \qquad \mathcal{V}(X)_{jk} = p(2\gamma-1)\lambda_j\lambda_kX_{jk} \quad (j \neq k).$$

Hence $\mathcal{T}$ maps the space $\mathbb{S}^{2P}_{\mathrm{diag}}$ of matrices whose four $P \times P$ blocks are diagonal in this basis into itself, and $\rho(\mathcal{T}) = \rho(\mathcal{T}_{\mathrm{diag}})$ for every $\eta > 0$, where $\mathcal{T}_{\mathrm{diag}} = \mathcal{T}|_{\mathbb{S}^{2P}_{\mathrm{diag}}}$.

In coordinates $(\mathbb{E}w_k^2, \mathbb{E}w_kw_k', \mathbb{E}w_k'^2)_{k\leq P}$ $(w' = w_{t-1})$, $\mathcal{T}_{\mathrm{diag}} = \mathrm{blkdiag}(\widetilde{B}_k) + \eta^2p\gamma\, rr^\top$ with $r = (\lambda_ke_1)_{k\leq P}$ and

$$\widetilde{B}_k = \begin{pmatrix} a_k^2 + \eta^2c_k & -2a_k\beta & \beta^2 \\ a_k & -\beta & 0 \\ 1 & 0 & 0 \end{pmatrix}.$$

Let g_k be the impulse response of mode k of deterministic heavy ball ($g_k(0) = 1$, $g_k(1) = a_k$, $g_k(\tau+1) = a_kg_k(\tau) - \beta g_k(\tau-1)$), $\psi_k(\tau) = \eta^2 g_k(\tau)^2$, $h_k(\tau) = \psi_k(\tau) + c_k\sum_{s<\tau}\psi_k(\tau-1-s)h_k(s)$, and

$$K_\tau = p\gamma\sum_k \lambda_k^2 h_k(\tau), \qquad \widehat{\psi}_k(z) = \frac{\eta^2(1+\beta z)}{(1-\beta z)\big(1-(a_k^2-2\beta)z+\beta^2z^2\big)},$$

$$\widehat{h}_k = \frac{\widehat{\psi}_k}{1 - c_kz\widehat{\psi}_k}, \qquad \widehat{K} = p\gamma\sum_k \lambda_k^2\widehat{h}_k.$$

$\widehat{\psi}_k$, $\widehat{h}_k$, $\widehat{K}$ are the generating functions of ψ_k, h_k, K (rational in z). Let $\sigma_k^2 = u_k^\top\Sigma_\xi u_k$ and

$$S(\eta) \triangleq \frac{\eta p\gamma}{2(1-\beta)}\sum_{k=1}^{P}\frac{\lambda_k}{1-\alpha_kx_k}.$$

Theorem 5 (Threshold, renewal equation, rate, prefactor, slowing down)**.** *In this model:*

(a) ***Threshold.*** $\rho(\mathcal{T}) < 1$ *if and only if* $\eta < \eta^\star$*, where* $\eta^\star$ *is the unique root of* $S(\eta) = 1$ *with* $\eta \max_k \max(\alpha_k, 0)\lambda_k < 2(1+\beta)$*; for* $\eta < \eta^\star$*,* $\ell_\infty \triangleq \operatorname{tr}(H\Sigma_0) = f_\infty/(1 - S(\eta))$ *with* $f_\infty = \sum_k \frac{\eta\sigma_k^2}{2(1-\beta)(1-\alpha_k x_k)}$*.*

(b) ***Renewal equation.*** *From* $w_0 = w_{-1} = 0$*,* $\ell_t = \operatorname{tr}(H\mathbb{E}w_t w_t^\top)$ *satisfies* $\ell_t = f_t + \sum_{s<t} K_{t-1-s}\ell_s$ *with* $f_t = \sum_{\tau<t}\phi_\tau$*,* $\phi_\tau = \sum_k \lambda_k\sigma_k^2 h_k(\tau)$*; equivalently* $\widehat{\ell}(z) = z\widehat{\phi}(z)/\big((1-z)(1-z\widehat{K}(z))\big)$*.*

(c) ***Rate.*** *If* $\rho(\mathcal{T}) > 0$ *(automatic for* $\beta > 0$*, since* $\rho(\mathcal{T}) \geq \rho(\bar{F})^2 \geq \beta$*), then* $\rho(\mathcal{T}) = 1/z_\star$*, where* $z_\star$ *is the smallest positive root of* $z\widehat{K}(z) = 1$*,* $\widehat{K}$ *being read as the rational function above (for* $c_k < 0$ *its power series may diverge at* $z_\star$*).*

(d) ***Single exponential.*** *If* $c_k \geq 0$ *for every* k *(i.e.* $\gamma(2+\kappa_k) \geq 1$*) and* $\eta < \eta^\star$*, then* $z_\star > 1$ *is the only root of* $z\widehat{K}(z) = 1$ *in* $|z| \leq z_\star$*, it is simple, and*

$$\ell_\infty - \ell_t = C\rho(\mathcal{T})^t + O(r^{-t}) \quad \text{for some } r > z_\star, \qquad C = \frac{\widehat{\phi}(z_\star)}{(z_\star - 1)\,\frac{d}{dz}\big(z\widehat{K}(z)\big)\big|_{z_\star}} > 0.$$

(e) ***Critical slowing down.*** *Under the condition of (d), as* $\eta \uparrow \eta^\star$*,*

$$1 - \rho(\mathcal{T}) = \frac{S'(\eta^\star)}{\sum_{\tau\geq 0}(\tau+1)K_\tau}\,(\eta^\star - \eta) + O\big((\eta^\star - \eta)^2\big), \qquad S'(\eta^\star) > 0,$$

the kernel being taken at $\eta^\star$*, where* $\sum_\tau K_\tau = S(\eta^\star) = 1$*; the denominator is the mean renewal time.*

The condition $c_k \geq 0$ is $\kappa_k \geq 1/\gamma - 2$; it holds for every $\kappa_k \geq -1$ (as $\gamma \geq 1$), and for Rademacher modes ($\kappa_k = -2$) it fails for every γ. For $c_k < 0$ the kernel can have both signs; (a)–(c) still hold, while h_k need not be summable (for $P = 1$, $\lambda_1 = 1$, $\beta = 0$, $B = 1$, Rademacher features and $\eta = 1.5$, $\widehat{h}_1 = 2.25/(1+2z)$, h_1 grows geometrically and $\rho(\mathcal{T}) = 1/4 = 1/z_\star$), and the single exponential of (d) can fail: for $P = 1$ Rademacher features $H_{B_t} \equiv \lambda_1$, $\mathcal{T}_{\text{diag}} = B_1$, all eigenvalues of B_1 have modulus β in the underdamped regime, and $(\ell_\infty - \ell_t)/\beta^t$ does not converge. From a general initial condition with $\mathbb{E}w_0 = \mathbb{E}w_{-1} = 0$ (or with $\mathbb{E}[\xi_t(H_{B_t} - H)_{jk}] = 0$ for all j, k), (b) holds with f_t replaced by $f_t + \sum_k \lambda_k \epsilon_k(t)$, where $\epsilon_k(t)$ propagates the initial 2×2 block of mode k through $\widetilde{B}_k$; otherwise a cross term $\eta^2\mathbb{E}[(H_{B_t} - H)\bar{w}_t\xi_t^\top]$ enters; without additive noise this is the interpolating setting of Lee et al. [2022] and Everett and Paquette [2026].

Corollary 6 (Special cases)**.** *(i) Gaussian features (*$\gamma = 1$*,* $\kappa_k \equiv 0$*):* $S(\eta) = 1$ *is the vanishing of the denominator* $1 - \frac{1}{B}\operatorname{tr}[\Lambda A G_\mu^{-1}]$ *of Ziyin et al. [2022, Theorem 2] (scalar learning rate) and the critical equation (56) of Wang et al. [2026].*

(ii) Per-sample gate on an arbitrary $H = q\tilde{H}$ *(*$\gamma = 1/q$*,* $\kappa_k = 0$*,* $\tilde{H} = \operatorname{diag}(\tilde{\lambda}_k)$*), in EMA form:*

$$\frac{\eta_{\text{EMA}}}{2B}\sum_{k=1}^{P}\frac{\tilde{\lambda}_k}{1 - \frac{\eta_{\text{EMA}}\tilde{\lambda}_k}{2}\left(q\frac{1-\beta}{1+\beta} + \frac{2-q}{B}\right)} = 1.$$

For $\tilde{H} = I_d$ *this is* $1/\eta^\star_{\text{EMA}} = \frac{d+2-q}{2B} + \frac{q}{2}\frac{1-\beta}{1+\beta}$*, equation (6) of Everett and Paquette [2026], and the rate equation of (c) reduces to a cubic; their Lemma H.4 gives the same per-step spectrum, written as* $1 + \varepsilon z$ *at* $\beta = 1 - \varepsilon$*.*

(iii) Per-coordinate gates: as $q_k \to 0$ at fixed $\tilde\lambda_k$, $\alpha_k\lambda_k \to p\kappa_\beta(3+\tilde\kappa_k)\tilde\lambda_k$, so in the limit of rare coordinates the admissible range of (a) becomes $\eta_{\mathrm{EMA}} < 2B/((3+\tilde\kappa_k)\tilde\lambda_k)$, which in EMA form does not depend on β (at finite q_k the range is $\eta_{\mathrm{EMA}}\tilde\lambda_k\big(q_k\frac{1-\beta}{1+\beta}+p(3+\tilde\kappa_k-2q_k)\big) < 2$, which does); all coordinates are coupled through the trace term of S. The gates here are independent across coordinates, unlike the exclusive tokens of Everett and Paquette [2026], whose rows decouple and whose noise ceiling $\eta < 2B/(d+2-p)$ [Everett and Paquette, 2026, Section 4.2] comes from the trace term, while this one comes from the diagonal self-term.

(iv) EMA form, momentum and batch. Write $\lambda_{\max} = \lambda_1$, $\lambda_{\min} = \lambda_P$, put $\theta = \frac{1-\beta}{1+\beta}$, $d_k = \frac{1}{2}\big(\theta\lambda_k + c_k/\lambda_k\big)$ and $M_\infty = \operatorname{diag}\big((\gamma(2+\kappa_k)-1)\lambda_k\big) + \gamma\sqrt{\lambda}\sqrt{\lambda}^\top$. With $\eta = (1-\beta)\eta_{\mathrm{EMA}}$,

$$S = \frac{\eta_{\mathrm{EMA}}\, p\gamma}{2}\sum_{k=1}^{P}\frac{\lambda_k}{1-\eta_{\mathrm{EMA}}d_k}, \qquad \frac{1}{\eta^\star_{\mathrm{EMA}}} = \lambda_{\max}\Big(\operatorname{diag}(d_k) + \frac{p\gamma}{2}\sqrt{\lambda}\sqrt{\lambda}^\top\Big),$$

and β enters only through θ. (a) For every level $s \in (0,1]$ the root $\eta_{\mathrm{EMA}}(\beta;s)$ of $S = s$ in the admissible range ($s = 1$ gives $\eta^\star_{\mathrm{EMA}}$) increases strictly with β, and

$$\frac{\beta}{1+\beta}\lambda_{\min} \le \frac{1}{\eta_{\mathrm{EMA}}(0;s)} - \frac{1}{\eta_{\mathrm{EMA}}(\beta;s)} \le \frac{\beta}{1+\beta}\lambda_{\max}.$$

(b) As $\beta \uparrow 1$, $1/\eta^\star_{\mathrm{EMA}}$ decreases to $\frac{p}{2}\lambda_{\max}(M_\infty)$, and $\lambda_{\max}(M_\infty) \ge \sqrt{\lambda_{(1)}\lambda_{(2)}} > 0$ for $P \ge 2$ ($\lambda_{(1)} \ge \lambda_{(2)}$ the two largest eigenvalues), so the EMA threshold stays bounded; for $s < 1$, $\eta_{\mathrm{EMA}}(\beta;s)$ increases to the finite root of $\frac{xp\gamma}{2}\sum_k \lambda_k/(1-xc_k/(2\lambda_k)) = s$. (c) With $\eta_{\mathrm{det}} = 2(1+\beta)/\lambda_{\max}$, $\eta^\star/\eta_{\mathrm{det}} = \lambda_{\max}/\lambda_{\max}\big(H + \frac{1+\beta}{(1-\beta)B}M_\infty\big)$ depends on (β, B) only through $B(1-\beta)/(1+\beta)$. If $\lambda_{\max}(M_\infty) > 0$, then $\eta^\star \ge \eta_{\mathrm{det}}/2$ if and only if $B \ge B_{1/2}(\beta) = \frac{1+\beta}{1-\beta}\lambda_{\max}(G^{-1/2}M_\infty G^{-1/2})$, $G = 2\lambda_{\max}I - H$ (in the eigenbasis of H), and $\lambda_{\max}(M_\infty)/(2\lambda_{\max}) \le B_{1/2}(0) \le \lambda_{\max}(M_\infty)/\lambda_{\max}$. For $\tilde H = I_d$ in (ii), (a)–(c) can be read from eq. (6) of Everett and Paquette [2026]; for spectrally expressible curvature noise, Gaussian features among them, the limit $\beta \to 1$ of the effective threshold is in Velikanov et al. [2023, eq. (23)]; mode-dependent κ_k is not of that form. Their critical batch size [Everett and Paquette, 2026, Proposition 2] counts samples and steps to a target loss and is a different quantity from $B_{1/2}$.

4 Two consequences

Proposition 7 (One-direction test against the true threshold)**.** *Let $\gamma = 1$, $\kappa_k \equiv \kappa$, $\alpha = 1+p\kappa_\beta(1+\kappa)$, $\mu^\star = 2(1+\beta)/\eta^\star$ and $\mu_{1\mathrm{D}} = \lambda_1\big(1+p\kappa_\beta(2+\kappa)\big)$, so that $2(1+\beta)/\mu_{1\mathrm{D}}$ is the one-dimensional threshold along u_1 [Andreyev et al., 2026, eq. (10)] with $\operatorname{Var}(u_1^\top H_{B_t}u_1) = p(2+\kappa)\lambda_1^2$. Then $\mu^\star \ge \mu_{1\mathrm{D}}$ and*

$$\alpha\,\frac{\operatorname{tr}H^2}{\operatorname{tr}H} \le \mu^\star - p\kappa_\beta \operatorname{tr}H \le \max(\alpha,0)\lambda_1.$$

In particular, at high effective rank $r_{\mathrm{eff}} = \operatorname{tr}H/\lambda_1$ the true threshold depends on κ only through a relative term $|(p\kappa_\beta)^{-1} + 1 + \kappa|/r_{\mathrm{eff}}$, while for Rademacher features ($\mu_{1\mathrm{D}} = \lambda_1$; $\alpha < 0$ when $B < \kappa_\beta$) the ratio of the one-dimensional threshold to the true one is at least $p\kappa_\beta r_{\mathrm{eff}} + \alpha\operatorname{tr}H^2/(\lambda_1\operatorname{tr}H)$.

Table 1: from Gaussian to Rademacher features the true threshold moves by 1.32% and the one-dimensional threshold by a factor 10.5; the bound of Proposition 7 gives a ratio of at least 144.460 at $\kappa = -2$ (actual 144.464). When $1+\kappa$ is of order r_{eff} or larger, the term $\alpha\lambda_1$ dominates and the kurtosis enters the true threshold at first order.

Table 1: $P = 256$, $\lambda_k = k^{-1/2}$ ($r_{\text{eff}} = 30.57$), $\beta = 0.9$, $B = 4$.

κ	-2 (Rademacher)	-1	0 (Gaussian)	2	10
true threshold $\eta^\star$	0.026304	0.026133	0.025961	0.025617	0.024223
one-dimensional threshold	3.8000	0.66087	0.36190	0.19000	0.065517
ratio	144.46	25.29	13.94	7.42	2.70

Proposition 8 (Fixed effective step)**.** *In the general setting of Section 2, let $\mathcal{V}$ be independent of β. (i) The effective threshold $\eta^\star(\beta)/(1-\beta)$ increases strictly with $\beta \in [0, 1)$, and $\frac{1}{\eta^\star(0)} - \frac{1-\beta}{\eta^\star(\beta)} \geq \frac{\beta}{1+\beta}\lambda_{\min}(H)$. (ii) Fix $\eta_{\text{eff}} > 0$ and let $\eta = (1-\beta)\eta_{\text{eff}}$. If $0 \leq \beta_1 < \beta_2 < 1$ are both stable, then $\operatorname{tr}(\Sigma_\xi \Sigma_0(\beta_2)) < \operatorname{tr}(\Sigma_\xi \Sigma_0(\beta_1))$. If Σ_ξ is proportional to H the loss $\operatorname{tr}(H\Sigma_0)$ therefore decreases strictly; in general it can increase: for $H = \operatorname{diag}(1, 0.3)$, two equiprobable minibatch Hessians $H \pm N$ with $N = \left(\begin{smallmatrix} -0.6 & -0.7 \\ -0.7 & -0.2 \end{smallmatrix}\right)$, $\Sigma_\xi = vv^\top + 10^{-3} I$, $v = (-0.8, 0.6)$ and $\eta_{\text{eff}} = 0.52$, raising β from 0.5 to 0.95 raises the loss from 0.34072 to 0.35659 and lowers $\operatorname{tr}(\Sigma_\xi \Sigma_0)$ from 0.43506 to 0.34534.*

At fixed small step, momentum SGD tracks SGD at step $\eta/(1-\beta)$ [Yuan et al., 2016]; Proposition 8 gives the sign of the departure at finite step, for the noise-weighted second moment.

Proposition 9 (Rate at a fixed loss floor)**.** *In the model of Section 3 let $\Sigma_\xi = \sigma^2 H/B$ (least squares with label noise of variance $\sigma^2 > 0$) and $\eta < \eta^\star$. Then $\ell_\infty = \frac{\sigma^2}{\gamma} \frac{S(\eta)}{1-S(\eta)}$, so a target floor fixes the level $s = \gamma\ell_\infty/(\sigma^2 + \gamma\ell_\infty)$ of S, and the EMA step that reaches it is $\eta_{\text{EMA}}(\beta; s)$ of Corollary 6(iv). If $c_k \geq 0$ for every k, then, with the mean renewal time $\bar{\tau} = \frac{1}{S}\sum_{\tau \geq 0}(\tau + 1)K_\tau$,*

$$\rho(\mathcal{T}) \geq S^{1/\bar{\tau}} \geq S, \qquad \text{so} \qquad 1 - \rho(\mathcal{T}) \leq 1 - S = \frac{\sigma^2}{\sigma^2 + \gamma\ell_\infty}.$$

The continuum, isotropic analogue of the last bound is $r_c < 1 - H$ in Everett and Paquette [2026, Lemma C.5(c)]. Table 2 maximizes $1 - \rho(\mathcal{T})$ over β along a fixed floor, with $\rho(\mathcal{T}) = 1/z_\star$ computed exactly. In one direction at a high floor momentum does not help; at a low floor the gain approaches 2 (rows 3 and 5), the factor that Everett and Paquette [2026, Proposition H.10] find for isotropic features; at a moderate floor with an anisotropic H it exceeds 2 (row 4), consistent with the conditioning gain that Everett and Paquette [2026, Proposition 2] find for a vocabulary at large batch. Numerically, along a fixed floor $(1 - \rho(\mathcal{T}))/(1-\beta) \to 1 - \eta_{\text{EMA}}(\beta; s)/\eta^\star_{\text{EMA}}(\beta)$ as $\beta \uparrow 1$ (in the fourth row, 0.61520 against 0.61516 at $\beta = 1 - 10^{-4}$); we do not prove this.

Table 2: Rate at a fixed level s of S (fixed floor), maximized over β (grid of 318 values in $[0, 1 - 10^{-4.5}]$, refined by golden-section search). Rows 4–6: $\lambda = (1, 0.5, 0.2, 0.05, 0.01)$, $B = 8$; row 6 has $\kappa = (5, 0, 2, 8, 1)$ and $\gamma = 2$.

features	s	$1 - \rho$ at $\beta = 0$	best β	$1 - \rho$ there	gain
$P = 1$, Gaussian, $B = 4$	0.3	0.53760	0	0.53760	1.000
$P = 1$, Gaussian, $B = 4$	0.01	0.14367	0.752	0.21215	1.477
$P = 1$, Gaussian, $B = 4$	0.001	0.015825	0.970	0.028811	1.821
$P = 5$, Gaussian	0.3	0.023537	0.897	0.060514	2.571
$P = 5$, Gaussian	0.01	0.0017510	0.996	0.0035060	2.003
$P = 5$, mixed κ_k, $\gamma = 2$	0.3	0.010833	0.942	0.016475	1.521

A Proofs

Lemma 1. (i) $\mathcal{T}(Q)^\top = \mathcal{T}(Q^\top)$; $\mathcal{T}^k$ is in Kraus form, hence positive, and the Russo–Dye theorem gives $\|\mathcal{T}^k\| = \|\mathcal{T}^k(I)\|$; Gelfand's formula then gives $\rho(\mathcal{T}) = \lim \|\mathcal{T}^k(I)\|^{1/k} \le \rho(\mathcal{T}|_{\mathbb{S}^{2P}})$, and the reverse inequality is trivial.

Let $m_t = \mathbb{E}s_t$ and $g_t = (-\eta\xi_t, 0)$. Since s_t is independent of (F_t, g_t) and $\mathbb{E}g_t = 0$, $m_{t+1} = \bar{F}m_t$ and $Q_{t+1} = \mathcal{T}(Q_t) + G + C_t$, with $C_t = \mathbb{E}[(F_t - \bar{F})m_t g_t^\top] + (\cdot)^\top$, $\|C_t\| \le c_0\|m_t\|$. Write $\mathcal{T} = \mathcal{T}_0 + \Delta$, $\mathcal{T}_0(Q) = \bar{F}Q\bar{F}^\top$, $\Delta(Q) = \mathrm{blkdiag}(\eta^2\mathcal{V}(Q_{11}), 0)$; both are positive, so $\rho(\bar{F})^2 = \rho(\mathcal{T}_0) \le \rho(\mathcal{T})$ [Berman and Plemmons, 1994].

(ii) If $\rho(\mathcal{T}) < 1$, then $\|m_t\| \le c_1 r^t$ with $r < 1$, and $Q_t = \mathcal{T}^t(Q_0) + \sum_{s<t}\mathcal{T}^{t-1-s}(G + C_s) \to \sum_k \mathcal{T}^k(G)$.

(iii) Let $\rho = \rho(\mathcal{T}) \ge 1$. By the Krein–Rutman theorem [Berman and Plemmons, 1994, Thm. 1.3.2] the adjoint $\mathcal{T}^*(Y) = \mathbb{E}[F_t^\top Y F_t]$ has an eigenmatrix $W \succeq 0$, $W \ne 0$, with eigenvalue ρ, and $W_{11} \ne 0$ (otherwise $W \succeq 0$ forces $W_{12} = 0$, and then $F_t^\top W F_t = \mathrm{blkdiag}(W_{22}, 0)$ forces $W = 0$), so $g_0 \triangleq \mathrm{tr}(W_{11}\Sigma_\xi) > 0$. The centered second moment $\widetilde{Q}_t = Q_t - m_t m_t^\top$ obeys $\widetilde{Q}_{t+1} = \mathcal{T}(\widetilde{Q}_t) + U_t$ with $U_t = \mathbb{E}[\upsilon_t\upsilon_t^\top] \succeq 0$, $\upsilon_t = (F_t - \bar{F})m_t + g_t$, because the centered state is independent of (F_t, g_t) and has mean zero. With $\bar{w}_t = \mathbb{E}w_t$ and $\Delta_t = H_{B_t} - H$, $\mathrm{tr}(WU_t) = \eta^2\varphi(\bar{w}_t)$, $\varphi(a) = \mathbb{E}[(\Delta_t a + \xi_t)^\top W_{11}(\Delta_t a + \xi_t)]$, and

$$\mathrm{tr}(W\widetilde{Q}_{t+1}) = \rho\,\mathrm{tr}(W\widetilde{Q}_t) + \eta^2\varphi(\bar{w}_t) \ge \mathrm{tr}(W\widetilde{Q}_t) + \eta^2\varphi(\bar{w}_t).$$

The parallelogram identity gives $\varphi(a) + \varphi(-a) = 2\mathbb{E}[a^\top\Delta_t W_{11}\Delta_t a] + 2g_0 \ge 2g_0$ for every a. Each 2×2 block of $\bar{F}$ in the eigenbasis of H has eigenvalues with product $\beta < 1$, and $\det(I - \bar{F}_k) = \eta\lambda_k > 0$; so the eigenvalues of modulus ≥ 1 are semisimple and the only one on the unit circle is -1 (a non-real pair on the circle would have product 1, and $+1$ is excluded by $\det(I - \bar{F}_k) > 0$). If m_0 has a component on the eigenvalues of modulus > 1, $\|m_t\|$ and hence $\mathrm{tr}\,Q_t$ grow geometrically. Otherwise $\bar{w}_t = (-1)^t a + o(1)$, so $\varphi(\bar{w}_{2k}) + \varphi(\bar{w}_{2k+1}) \ge g_0$ for large k, and $\mathrm{tr}(W\widetilde{Q}_t)$ grows at least linearly; $\mathrm{tr}\,Q_t \ge \mathrm{tr}(W\widetilde{Q}_t)/\|W\|$. Finally $\mathbb{E}\|w_t\|^2 + \mathbb{E}\|w_{t-1}\|^2 > 0$ for every $t \ge 1$: if $w_t = w_{t-1} = 0$ a.s., then $\eta\xi_{t-1} = -\beta w_{t-2}$ a.s. with w_{t-2} independent of ξ_{t-1}, so ξ_{t-1} is a.s. constant, contradicting $\Sigma_\xi \succ 0$. The bound ct follows. The counterexample with $H = 0$ satisfies $(H_{B_t} - H)w_t + \xi_t = 0$ on the path $w \equiv 1$.

Theorem 2. By Lemma 1(ii), $\mathbb{E}w_t = 0$ in the stationary regime, and $(H_{B_t} - H, \xi_t)$ is independent of (w_t, w_{t-1}), so $\mathbb{E}[(H_{B_t} - H)w_t\xi_t^\top] = 0$ whatever the joint law of (H_{B_t}, ξ_t). With $A = (1+\beta)I - \eta H$ and $\Sigma_1 = \mathbb{E}w_{t+1}w_t^\top$: conditioning on the past gives $\Sigma_1 = A\Sigma_0 - \beta\Sigma_1^\top$, hence $\Sigma_1 = (A\Sigma_0 - \beta\Sigma_0 A)/(1-\beta^2)$ and $\Sigma_1 + \Sigma_1^\top = 2\Sigma_0 - \frac{\eta}{1+\beta}(H\Sigma_0 + \Sigma_0 H)$, which is (1) since $D = 2\Sigma_0 - \Sigma_1 - \Sigma_1^\top$. Expanding $\Sigma_0 = \mathbb{E}w_{t+1}w_{t+1}^\top$ gives $\Sigma_0 = A\Sigma_0 A + \beta^2\Sigma_0 - \beta(A\Sigma_1 + \Sigma_1^\top A) + \eta^2\mathcal{V}(\Sigma_0) + \eta^2\Sigma_\xi$; with $A^2\Sigma_0 + \Sigma_0 A^2 - 2\beta A\Sigma_0 A = [A,[A,\Sigma_0]] + 2(1-\beta)A\Sigma_0 A$ and $[A,[A,\Sigma_0]] = \eta^2[H,[H,\Sigma_0]]$ this is $(1-\beta^2)\Sigma_0 - \frac{1-\beta}{1+\beta}A\Sigma_0 A = -\frac{\beta\eta^2}{1-\beta^2}[H,[H,\Sigma_0]] + \eta^2\mathcal{V}(\Sigma_0) + \eta^2\Sigma_\xi$; expanding $A\Sigma_0 A$ and dividing by $\eta(1-\beta)$ gives (2).

Lemma 4. $\mathcal{V}(X) = p(\mathbb{E}[zz^\top Xzz^\top] - HXH)$ and $\mathbb{E}[z_j z_l z_m z_k] = \gamma\sqrt{\lambda_j\lambda_l\lambda_m\lambda_k}\,\mathbb{E}[\omega_j\omega_l\omega_m\omega_k]$, which vanishes unless every index appears an even number of times. For $j = k$: $l = m \ne k$ gives $\gamma\lambda_k\lambda_l X_{ll}$ and $l = m = k$ gives $\gamma(3+\kappa_k)\lambda_k^2 X_{kk}$. For $j \ne k$: $(l,m) = (j,k)$ and (k,j) each give $\gamma\lambda_j\lambda_k X_{jk}$ on symmetric X. Subtracting $(HXH)_{jk} = \lambda_j\lambda_k X_{jk}$ gives the formulas. Thus $\mathcal{V}$ maps diagonal matrices to diagonal ones and off-diagonal entries to themselves, and $\bar{F}$ acts mode by mode, so $\mathcal{T}(\mathbb{S}^{2P}_{\mathrm{diag}}) \subseteq \mathbb{S}^{2P}_{\mathrm{diag}}$. Since $I \in \mathbb{S}^{2P}_{\mathrm{diag}}$, $\|\mathcal{T}^k\| = \|\mathcal{T}^k(I)\| = \|\mathcal{T}^k_{\mathrm{diag}}(I)\| \le \|\mathcal{T}^k_{\mathrm{diag}}\| \le \|\mathcal{T}^k\|$ by Lemma 1(i), and Gelfand's formula gives $\rho(\mathcal{T}) = \rho(\mathcal{T}_{\mathrm{diag}})$.

Theorem 5. *Generating functions.* The mean recursion of mode k started at $(w, w') = (1, 0)$ is g_k, so $e_1^\top B_k^\tau e_1 = g_k(\tau)^2$ for $B_k = \widetilde{B}_k - \eta^2 c_k e_1 e_1^\top$, and $e_1^\top \widetilde{B}_k^\tau e_1 = h_k(\tau)/\eta^2$ by expanding the rank-

one perturbation. From $g_k(\tau) = \frac{r_+^{\tau+1} - r_-^{\tau+1}}{r_+ - r_-}$ with $r_+ + r_- = a_k$, $r_+ r_- = \beta$, summing the geometric series gives $\widehat{\psi}_k$; the three poles $1/\beta$, $r_\pm^{-2}$ are the reciprocals of the eigenvalues $\beta, r_\pm^2$ of B_k. At $z = 1$, $\widehat{\psi}_k(1) = \frac{\eta}{2(1-\beta)\lambda_k(1-x_k)}$ and $\widehat{h}_k(1) = \frac{\eta}{2(1-\beta)\lambda_k(1-\alpha_k x_k)}$, because $(1-x_k)(1-c_k\widehat{\psi}_k(1)) = 1-\alpha_k x_k$. Hence $p\gamma \sum_k \lambda_k^2 \widehat{h}_k(1) = S(\eta)$ whenever no pole lies at 1.

(b). From $Q_0 = 0$, $Q_t = \sum_{s<t} \mathcal{T}_0^{t-1-s}(\Delta(Q_s) + G)$ and Lemma 4 give $x_{t,k} = \mathbb{E}w_{t,k}^2 = \sum_{s<t} \psi_k(t-1-s)\big(\sigma_k^2 + p\gamma\lambda_k \ell_s + c_k x_{s,k}\big)$, whose solution is $x_{t,k} = \sum_{s<t} h_k(t-1-s)(\sigma_k^2 + p\gamma\lambda_k\ell_s)$; multiplying by λ_k and summing gives the equation, and generating functions give $\widehat{\ell} = \frac{z}{1-z}\widehat{\phi} + z\widehat{K}\widehat{\ell}$.

(a). The matrix $\mathcal{M} = \mathrm{diag}(\alpha_k\lambda_k) + p\kappa_\beta\gamma\lambda\mathbf{1}^\top$ is similar to $\widetilde{\mathcal{M}} = \mathrm{diag}(\alpha_k\lambda_k) + p\kappa_\beta\gamma\sqrt{\lambda}\sqrt{\lambda}^\top$, whose diagonal entries $\lambda_k(1 + p\kappa_\beta(\gamma(3+\kappa_k)-1)) \geq \lambda_k$ and off-diagonal entries are positive. The determinant lemma, $\det(\mu - \widetilde{\mathcal{M}}) = \prod_k(\mu - \alpha_k\lambda_k)\big(1 - p\kappa_\beta\gamma\sum_k \frac{\lambda_k}{\mu - \alpha_k\lambda_k}\big)$, the Perron vector of $\widetilde{\mathcal{M}} + cI$ (c large), which is positive and so excludes $\mu^\star = \alpha_k\lambda_k$, and the strict decrease of the sum in μ show that $\mu^\star = \lambda_{\max}(\widetilde{\mathcal{M}})$ is the unique root with $\mu > \max(0, \max_k \alpha_k\lambda_k)$; with $\mu = 2(1+\beta)/\eta$ the sum equals $S(\eta)$. By Lemma 4, $\rho(\mathcal{T}) = \rho(\mathcal{T}_{\mathrm{diag}})$, and the stationary equations of $\mathcal{T}_{\mathrm{diag}}$ with Σ_ξ replaced by $\mathrm{diag}(\sigma_k^2)$ ($\rho(\mathcal{T})$ does not depend on Σ_ξ) are the diagonal entries of Theorem 2(2), i.e. $(I - N)c' = b$ for $c'_k = 2\lambda_k\mathbb{E}w_k^2/(\eta(1+\beta))$, $N = \frac{\eta}{2(1+\beta)}\mathcal{M} \geq 0$ and $b = \sigma^2/(1-\beta^2) > 0$. If $\rho(\mathcal{T}) < 1$, a stationary solution exists with $c' \geq b > 0$ and $Nc' < c'$, so $\rho(N) < 1$ by the Collatz–Wielandt formula, i.e. $\eta < \eta^\star$. Conversely, for $\eta < \eta^\star$, $I - N$ is a nonsingular M-matrix and $c' > 0$; with $\Sigma_0 = \mathrm{diag}(\mathbb{E}w_k^2)$ and $\Sigma_1 = A\Sigma_0/(1+\beta)$ the matrix $Q = \left(\begin{smallmatrix} \Sigma_0 & \Sigma_1 \\ \Sigma_1 & \Sigma_0 \end{smallmatrix}\right)$ is positive definite ($|1 - 2x_k| < 1$ because $\eta < \eta^\star \leq 2(1+\beta)/\lambda_1$, as $\mu^\star \geq e_1^\top \widetilde{\mathcal{M}} e_1 \geq \lambda_1$) and solves $Q = \mathcal{T}(Q) + G$; then $Q_\varepsilon = Q - \varepsilon G \succ 0$ and $Q_\varepsilon - \mathcal{T}(Q_\varepsilon) = (1-\varepsilon)G + \varepsilon\mathcal{T}(G) \succ 0$ for small $\varepsilon > 0$, which forces $\rho(\mathcal{T}) < 1$. The formula for ℓ_∞ follows from $\mathbb{E}w_k^2 = \widehat{h}_k(1)(\sigma_k^2 + p\gamma\lambda_k\ell_\infty)$.

(c). Write $\mathcal{T}_{\mathrm{diag}} = \mathcal{A} + \eta^2 p\gamma r r^\top$, $\mathcal{A} = \mathrm{blkdiag}(\widetilde{B}_k)$; then $\eta^2 p\gamma\, r^\top(I - z\mathcal{A})^{-1} r = \widehat{K}(z)$ and $\det(I - z\mathcal{T}_{\mathrm{diag}}) = \prod_k \det(I - z\widetilde{B}_k)\big(1 - z\widehat{K}(z)\big)$. $\mathcal{T}_{\mathrm{diag}}$ preserves the cone of matrices whose P blocks $\left(\begin{smallmatrix} x_k & y_k \\ y_k & m_k \end{smallmatrix}\right)$ are positive semidefinite, so $\rho = \rho(\mathcal{T}_{\mathrm{diag}}) > 0$ has an eigenvector $X \neq 0$ in this cone. If $r^\top X = \sum_k \lambda_k x_k = 0$, then every $x_k = 0$, hence $y_k = 0$, and the third coordinate of $\mathcal{T}_{\mathrm{diag}} X$ is $x_k = 0 = \rho m_k$, so $X = 0$; thus $r^\top X > 0$ and $(\rho - \mathcal{A})X = \eta^2 p\gamma (r^\top X) r \neq 0$. If ρ were an eigenvalue of $\widetilde{B}_k$, the $(1,1)$ cofactor of $\rho - \widetilde{B}_k$, which equals $\rho(\rho + \beta) > 0$ for every c_k, would make the adjugate nonzero with a left null vector ℓ, $\ell_1 \neq 0$, so $e_1 \notin \mathrm{range}(\rho - \widetilde{B}_k)$, a contradiction. Hence $X = \eta^2 p\gamma (r^\top X)(\rho - \mathcal{A})^{-1} r$ and $z\widehat{K}(z) = 1$ at $z = 1/\rho$. Any positive root z' of $z\widehat{K} = 1$ at which $\widehat{K}$ is finite makes the determinant vanish, so $1/z' \leq \rho$.

(d). With $c_k \geq 0$, $h_k \geq 0$, $K_\tau \geq 0$ and $K_0 = p\gamma\eta^2 \sum_k \lambda_k^2 > 0$. Each $\widehat{h}_k$ is rational with nonnegative coefficients, so its smallest-modulus singularity is a positive pole where it tends to $+\infty$ (Pringsheim), or it is a polynomial; hence $z\widehat{K}$ increases from 0 to $+\infty$ on $[0, R_K)$, $R_K = \min_k R_k$, and has a unique root there, which is $z_\star = 1/\rho$ by (c), and $z_\star > 1$ since $\rho < 1$. For $|z| \leq z_\star$, $z \neq z_\star$: $\mathrm{Re}(z\widehat{K}(z)) = \sum_\tau K_\tau \,\mathrm{Re}\, z^{\tau+1} < \sum_\tau K_\tau z_\star^{\tau+1} = 1$, since equality would need $z^{\tau+1} > 0$ for $\tau = 0$. So $1 - z\widehat{K}$ has no other zero in $|z| \leq z_\star$; the zero is simple as $\frac{d}{dz}(z\widehat{K})(z_\star) = \sum_\tau (\tau+1)K_\tau z_\star^\tau > 0$; and the poles of $\widehat{\phi}$ lie in $|z| \geq R_K > z_\star$. The rational function $\widehat{\ell}$ thus has, in $|z| \leq z_\star$, only simple poles at 1 (residue $-\ell_\infty$) and at $z_\star$, which gives the expansion with the stated C; $C > 0$ because $\widehat{\phi}(z_\star) > 0$, $z_\star > 1$ and the derivative is positive.

(e). At $\eta^\star$, $\rho(\mathcal{T}) = 1$ by continuity, so $z_\star = 1$; and $1 < R_K$: $c_k \geq 0$ gives $\alpha_k \geq 1$, the admissible range gives $\alpha_k x_k < 1$ and hence $x_k < 1$, so the poles of $\widehat{\psi}_k$ lie outside the closed unit disk and $c_k z\widehat{\psi}_k(z) \leq c_k\widehat{\psi}_k(1) < 1$ on $[0,1]$, the last inequality being $(1 - x_k)(1 - c_k\widehat{\psi}_k(1)) = 1 - \alpha_k x_k > 0$; hence $R_k > 1$. The argument of (d) applies at $\eta^\star$ as well, so $z_\star(\eta)$ is the branch through $(1, \eta^\star)$. Thus $\Phi(z, \eta) = z\widehat{K}(z;\eta)$ is analytic near $(1, \eta^\star)$, with $\partial_z\Phi = \sum_\tau(\tau+1)K_\tau > 0$ and $\partial_\eta\Phi = S'(\eta^\star)$ because $\widehat{K}(1;\eta) = S(\eta)$ there. Each term of S has the form $\eta/(1 - \eta d_k)$ with $\eta d_k < 1$, so $S' > 0$. The implicit function theorem gives $z_\star(\eta)$ with $z_\star'(\eta^\star) = -S'/\partial_z\Phi$, and $\rho = 1/z_\star$.

Corollary 6. (i) With $\Lambda = \eta I$, $A = H$, $S = B$ and $\mu = \beta$, $\frac{1}{B}\operatorname{tr}[\Lambda A G_\mu^{-1}] = \frac{\eta}{B}\sum_k \frac{\lambda_k}{2(1-\beta)-(\frac{1-\beta}{1+\beta}+\frac{1}{B})\eta\lambda_k}$, which is $S(\eta)$ at $\gamma = 1$, $\kappa_k = 0$ ($\alpha_k = 1 + p\kappa_\beta$); equation (56) of Wang et al. [2026] is the same sum. (ii) Substitute $\lambda_k = q\tilde{\lambda}_k$, $\gamma = 1/q$, $\kappa_k = 0$, $\eta = (1-\beta)\eta_{\mathrm{EMA}}$ in S; for $\tilde{\lambda}_k \equiv 1$ the equation is linear in η_{EMA}, and in (c) all modes are equal, so $1 = z\widehat{K}(z)$ becomes $Q_3(z) = \eta^2(Pp\gamma\lambda^2 + c)z(1+\beta z)$ with Q_3 the cubic denominator of $\widehat{\psi}$. (iii) $\alpha_k\lambda_k = q_k\tilde{\lambda}_k + p\kappa_\beta\tilde{\lambda}_k(3 + \tilde{\kappa}_k - 2q_k)$. (iv) From $\alpha_k\lambda_k = \lambda_k + \kappa_\beta c_k/\lambda_k$, $\alpha_k x_k = \eta_{\mathrm{EMA}} d_k$ and $\frac{\eta p\gamma}{2(1-\beta)} = \frac{\eta_{\mathrm{EMA}} p\gamma}{2}$; with $\mu^\star = \lambda_{\max}(\widetilde{\mathcal{M}})$ from the proof of Theorem 5(a), $1/\eta^\star_{\mathrm{EMA}} = \mu^\star/(2\kappa_\beta)$ gives the second formula, and $\widetilde{\mathcal{M}} = H + p\kappa_\beta M_\infty$ gives (c). (a) With $y = 1/\eta_{\mathrm{EMA}}$, the level s is the root of $F_\theta(y) = \sum_k \lambda_k/(y - d_k) = 2s/(p\gamma)$ on $y > \max(\max_k d_k, 0)$, where F_θ decreases; it exists for $s \le 1$ because it exists for $s = 1$. From $\beta = 0$ to β every d_k decreases by $\delta_k = \frac{\beta}{1+\beta}\lambda_k$. If y_0 is the root at $\beta = 0$, then $y_0 \ge \mu^\star/2 \ge \lambda_{\max}/2 > \max_k \delta_k$, $F_\theta(y_0 - \min_k \delta_k) \le F_1(y_0)$ and $F_\theta(y_0 - \max_k \delta_k) \ge F_1(y_0)$ (when $y_0 - \max_k \delta_k$ is admissible), so the root at β lies in $[y_0 - \max_k \delta_k,\, y_0 - \min_k \delta_k]$; the same comparison between $\beta_1 < \beta_2$ gives the strict increase. (b) As $\theta \downarrow 0$, $\operatorname{diag}(d_k)$ decreases to $\operatorname{diag}(c_k/(2\lambda_k)) = \frac{p}{2}\operatorname{diag}((\gamma(2+\kappa_k)-1)\lambda_k)$. Since $(\gamma(2+\kappa_k)-1)\lambda_k \ge -\lambda_k$ and $\gamma \ge 1$, $M_\infty \succeq -H + \sqrt{\lambda}\sqrt{\lambda}^\top$, whose principal 2×2 submatrix on the two largest eigenvalues has top eigenvalue $\sqrt{\lambda_{(1)}\lambda_{(2)}}$; Cauchy interlacing gives the bound. For $s < 1$ the limiting root is finite: a mode with $c_k > 0$ gives a pole, a mode with $c_k = 0$ an unbounded term, and if every $c_k < 0$ the limiting sum tends to $\sum_k \gamma/(1-\gamma(2+\kappa_k)) \ge P\gamma \ge 1 > s$. (c) Since $G \succ 0$, $\lambda_{\max}(H + \tau M_\infty) < 2\lambda_{\max}$ if and only if $\tau G^{-1/2} M_\infty G^{-1/2} \prec I$; with $\tau = \frac{1+\beta}{(1-\beta)B}$ this is $B > B_{1/2}(\beta)$, and equality on one side is equality on the other. Ostrowski's theorem with $\lambda_{\max} I \preceq G \preceq 2\lambda_{\max} I$ gives the bracket.

Proposition 7. Here $\widetilde{\mathcal{M}} = \alpha H + p\kappa_\beta\sqrt{\lambda}\sqrt{\lambda}^\top$ in the eigenbasis of H and $\mu^\star = \lambda_{\max}(\widetilde{\mathcal{M}})$. The Rayleigh quotients at e_1 and at $\sqrt{\lambda}/\sqrt{\operatorname{tr} H}$ give $\mu^\star \ge (\alpha + p\kappa_\beta)\lambda_1 = \mu_{1\mathrm{D}}$ and $\mu^\star \ge \alpha \operatorname{tr} H^2/\operatorname{tr} H + p\kappa_\beta \operatorname{tr} H$; Weyl's inequality gives $\mu^\star \le \max_k \alpha\lambda_k + p\kappa_\beta \operatorname{tr} H \le \max(\alpha, 0)\lambda_1 + p\kappa_\beta \operatorname{tr} H$. The ratio of thresholds is $\mu^\star/\mu_{1\mathrm{D}}$.

Proposition 9. Here $\sigma_k^2 = \sigma^2 p\lambda_k$, so $f_\infty = \frac{\sigma^2}{\gamma} S$ and Theorem 5(a) gives ℓ_∞. With $c_k \ge 0$, $K_\tau \ge 0$, $\sum_\tau K_\tau = \widehat{K}(1) = S$ and $\sum_\tau K_\tau z_\star^{\tau+1} = 1$ with $z_\star = 1/\rho(\mathcal{T}) > 1$; by convexity of $x \mapsto z_\star^x$, $1/S = \sum_\tau \frac{K_\tau}{S} z_\star^{\tau+1} \ge z_\star^{\bar{\tau}}$, so $\rho(\mathcal{T}) \ge S^{1/\bar{\tau}}$, and $\bar{\tau} \ge 1$, $S < 1$ give $S^{1/\bar{\tau}} \ge S$. The table is part G of `note-v2-checks.py`.

Proposition 8. With $\eta = (1-\beta)\eta_{\mathrm{eff}}$, Theorem 2(2) reads $M_\beta\Sigma_0 = \eta_{\mathrm{eff}}\Sigma_\xi$, $M_\beta = \mathcal{L}_0 - \eta_{\mathrm{eff}}\mathcal{A}_\beta$, $\mathcal{A}_\beta = (1-\beta)\mathcal{B}_\beta$, and $M_\beta \succ 0$ at stable β by Proposition 3. In the eigenbasis of H, $\mathcal{A}_0 - \mathcal{A}_\beta$ multiplies entry (j,k) by $m_{jk}(\beta) = \frac{2\beta}{1+\beta}\lambda_j\lambda_k + \frac{\beta}{1-\beta^2}(\lambda_j - \lambda_k)^2$, which increases strictly in β and is positive. (i) By Proposition 3, $1/\eta^\star_{\mathrm{eff}}(\beta) = \lambda_{\max}(\mathcal{L}_0^{-1/2}\mathcal{A}_\beta\mathcal{L}_0^{-1/2})$, and $\mathcal{L}_0^{-1/2}(\mathcal{A}_{\beta_1} - \mathcal{A}_{\beta_2})\mathcal{L}_0^{-1/2} \succ 0$ for $\beta_1 < \beta_2$, so Weyl's inequality gives the strict increase; $m_{jk}(\beta) \ge \frac{\beta}{1+\beta}(\lambda_j^2 + \lambda_k^2) \ge \frac{\beta}{1+\beta}\lambda_{\min}(H)(\lambda_j + \lambda_k)$ gives $\mathcal{L}_0^{-1/2}(\mathcal{A}_0 - \mathcal{A}_\beta)\mathcal{L}_0^{-1/2} \succeq \frac{\beta}{1+\beta}\lambda_{\min}(H) I$ and the margin. (ii) $M_{\beta_2} - M_{\beta_1} \succ 0$, $M_{\beta_2}^{-1} \prec M_{\beta_1}^{-1}$, and $\operatorname{tr}(\Sigma_\xi\Sigma_0) = \eta_{\mathrm{eff}}\langle \Sigma_\xi, M_\beta^{-1}\Sigma_\xi\rangle$ decreases strictly. The example is computed in `note-v2-checks.py` (part E), where both values of β have $\rho(\mathcal{T}) < 1$ (0.809 and 0.966); the minibatch Hessian $H + N$ is indefinite (eigenvalues -0.466, 0.966) and $H - N$ is positive definite.

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